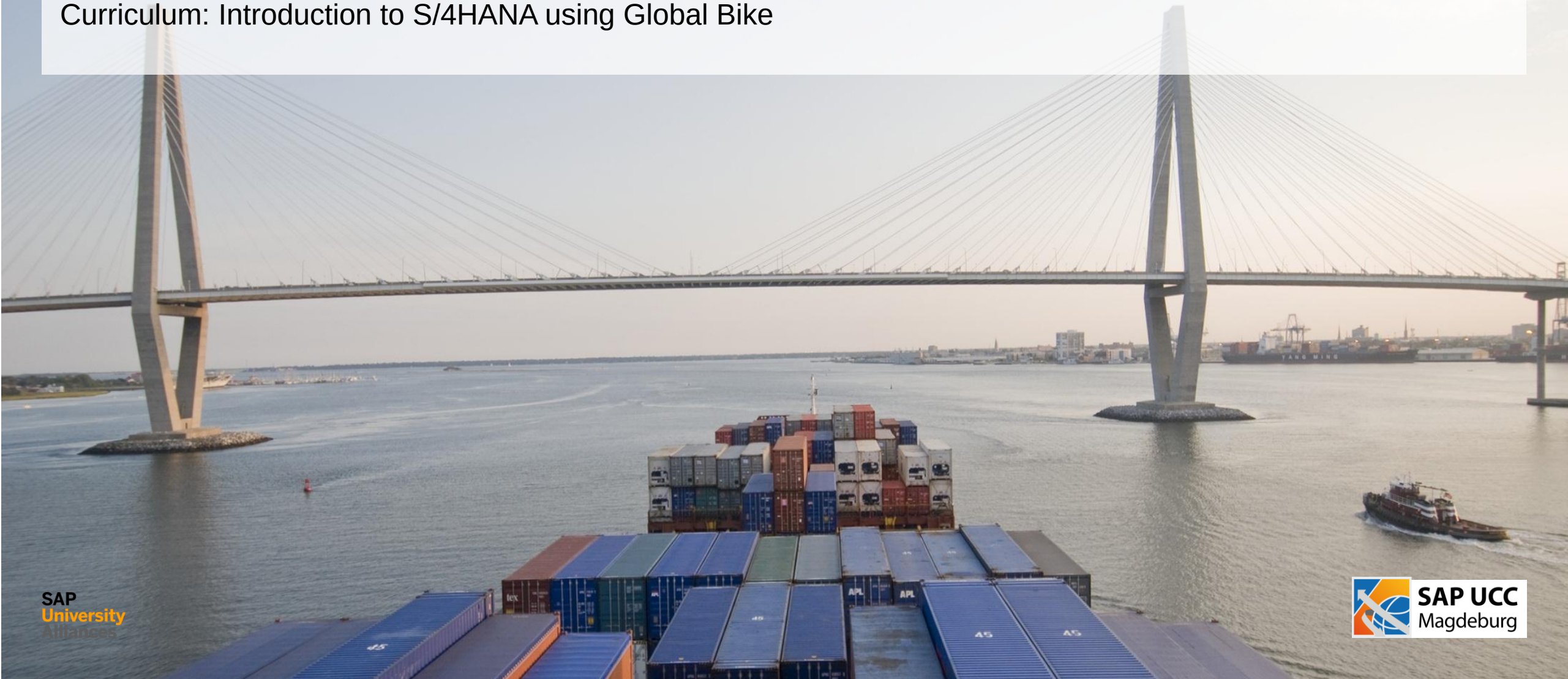
Warehouse Management (WM)

Curriculum: Introduction to S/4HANA using Global Bike



SAP University Alliances

SAP UCC Magdeburg

Teaching material - Information

Teaching material - Version

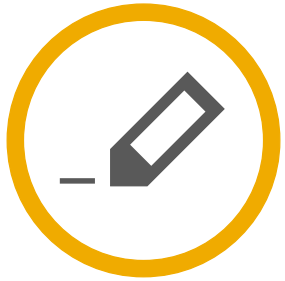
- 4.3 (July 2025)

- Software used
  - SAP S/4HANA 2023
  - Fiori 3.0
- Model
  - Global Bike

- Prerequisites
  - No Prerequisites needed

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Module Information



Authors

- Chris Bernhardt
- Babett Ruß



Target Audience

- Beginner

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Module Information



Learning Objectives

You are able to

- name some functionalities of the WM module.
- define the central organizational structures of the WM module.
- summarize the master data which is most important for the WM module.
- explain standard Warehouse Management processes.

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Functionality

- Goods Receipt
- Goods Issue
- Picking
- Packing
- Shipping
- Physical Inventory

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Agenda

- WM Organizational Structure
- WM Master Data
- WM Process Management and Control

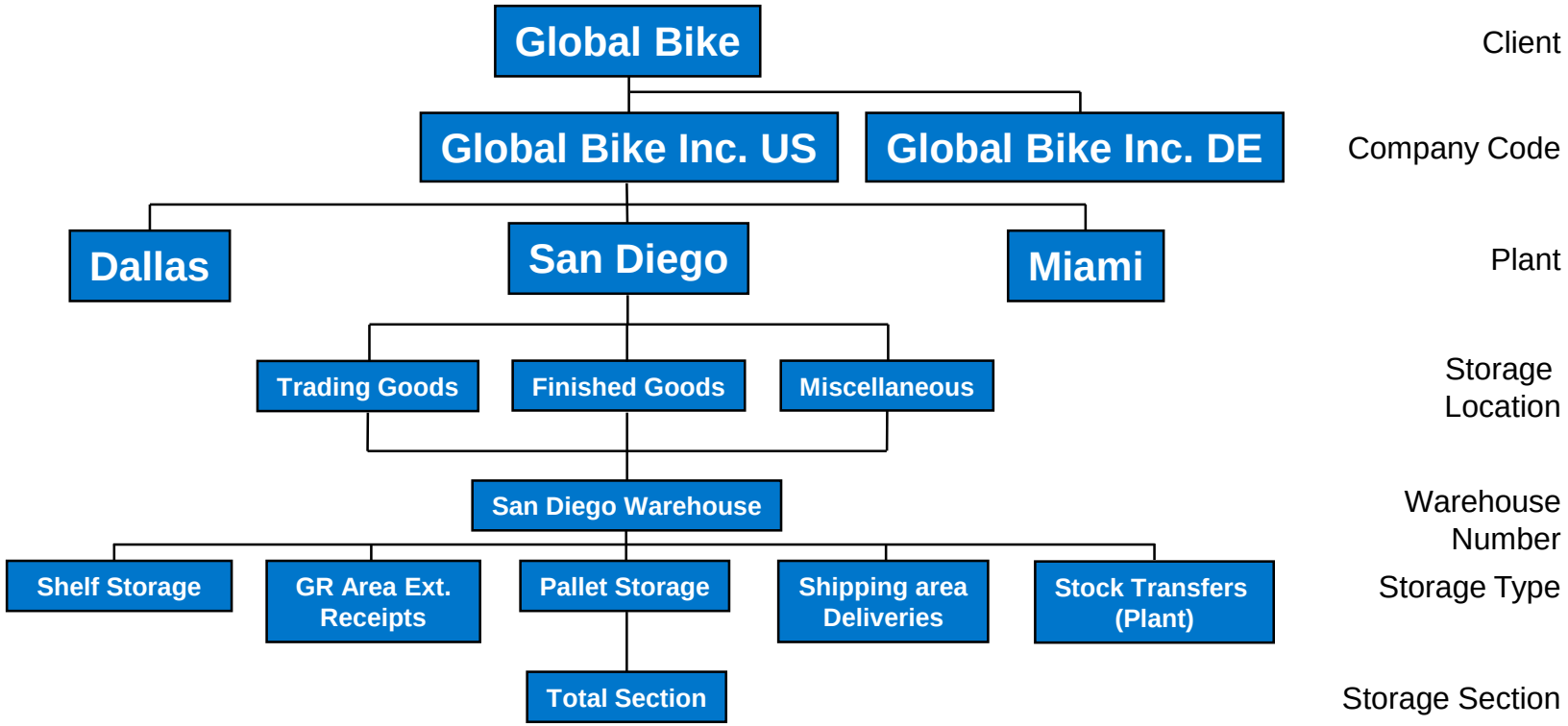
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Agenda

- WM Organizational Structure
- WM Master Data
- WM Process Management and Control

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WM Organizational Structure



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WM Organizational Structure

- Client
  - An independent environment in the system
- Company Code
  - Smallest organisational unit for which you can maintain a legal set of books
- Plant
  - Operating area or branch within a company
    - Manufacturing, distribution, purchasing or maintenance facility
- Storage Location
  - An organizational unit allowing differentiation between the various stocks of a material in a plant

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WM Organizational Structure

- Warehouse Number
  - Number that identifies a complex, physical warehouse structure within the warehouse management system
- Storage Type
  - Subdivision of a complex, physical warehouse
  - Is identified by its warehousing technique, form of organization, or its function
- Storage Section
  - Logical or physical subdivision of a storage type
  - Groups together a series of similar storage bins
- Picking Area
  - Groups storage bins together from the standpoint of picking strategies

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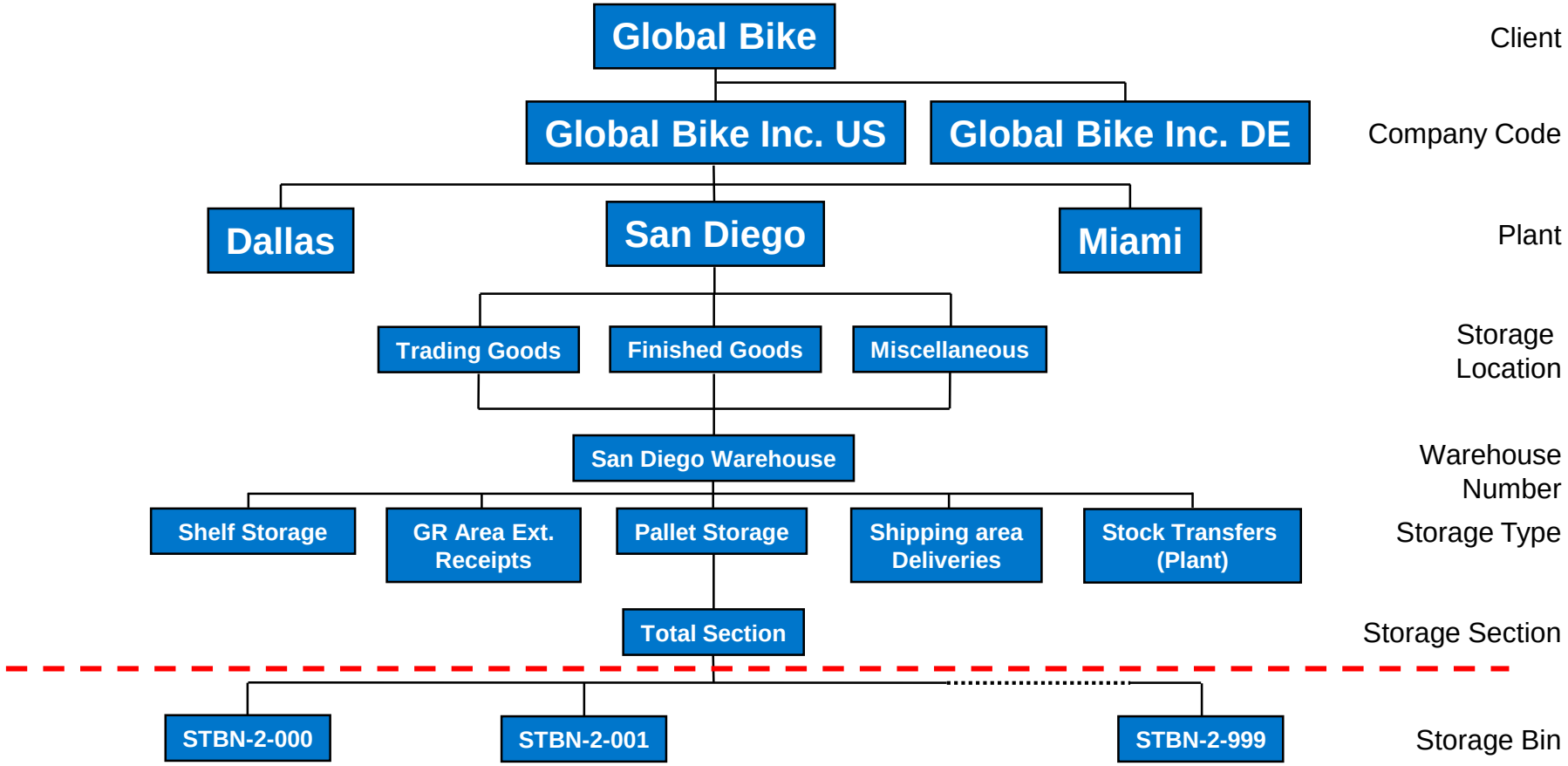
WM Organizational Structure

- Warehouse door
  - Marks the point where the means of transport changes to an internal transportation
- Allocation Zone
  - Temporary storage of goods, which have just arrived or will be shipped soon
- Shipping Point
  - Is the part of the company responsible for the type of shipping, the necessary shipping materials and the means of transport
- Storage Bin
  - Smallest addressable unit in a warehouse, which identifies the exact location in the warehouse where goods can be stored

→ Master Data

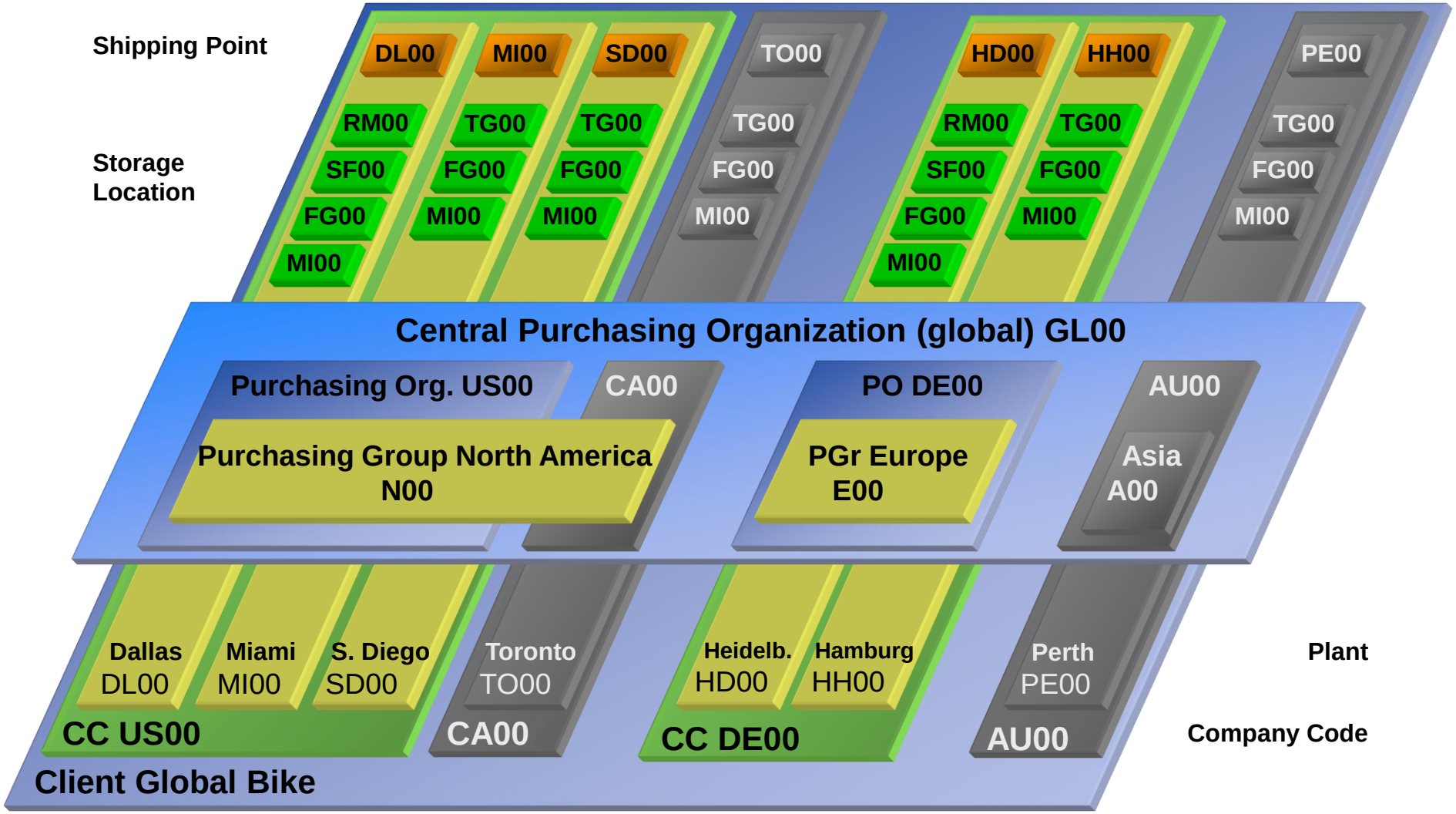
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Global Bike Structure for Warehouse Management



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Global Bike Enterprise Structure in SAP ERP (Logistics)



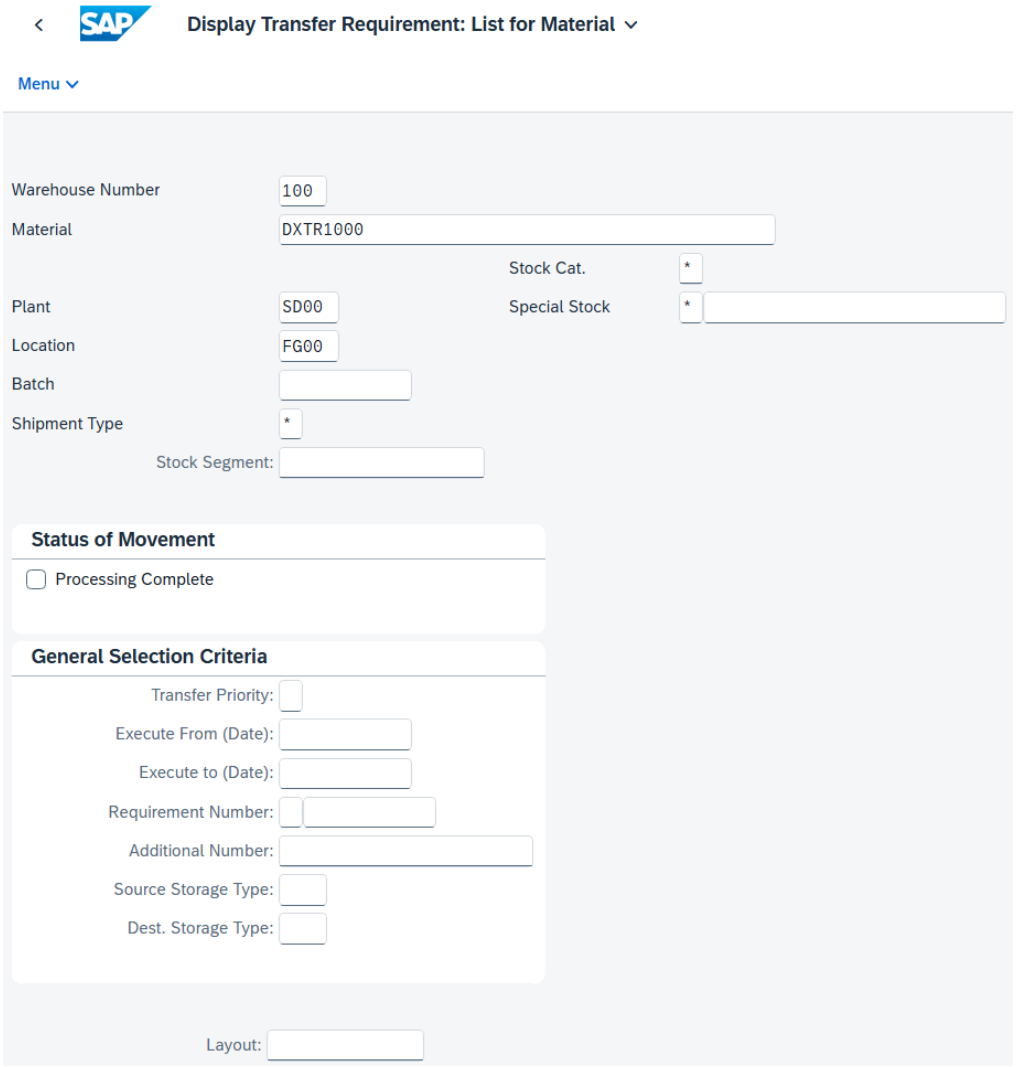
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Agenda

- WM Organizational Structure
- WM Master Data
- WM Process Management and Control

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WM Master Data



Storage Bin Master Data

Material Master Data

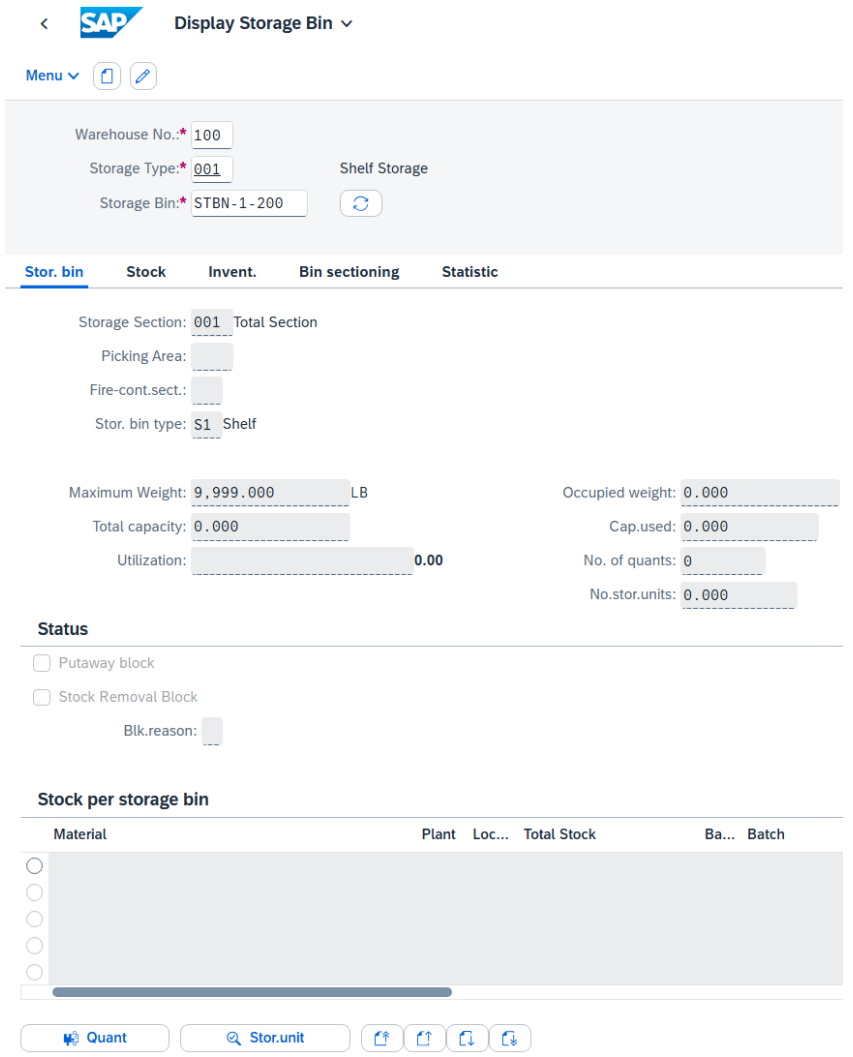
Hazard Master Data

Batch Master Data

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Storage Bin Master Data

- Storage Bin Master Data
  - Two different organizational levels
    - Warehouse Level:
      All indicators and fields which are valid for the whole warehouse
    - Storage Type Level:
      All indicators which are only valid for specific storage types
  - Also includes Storage Bins and Quants



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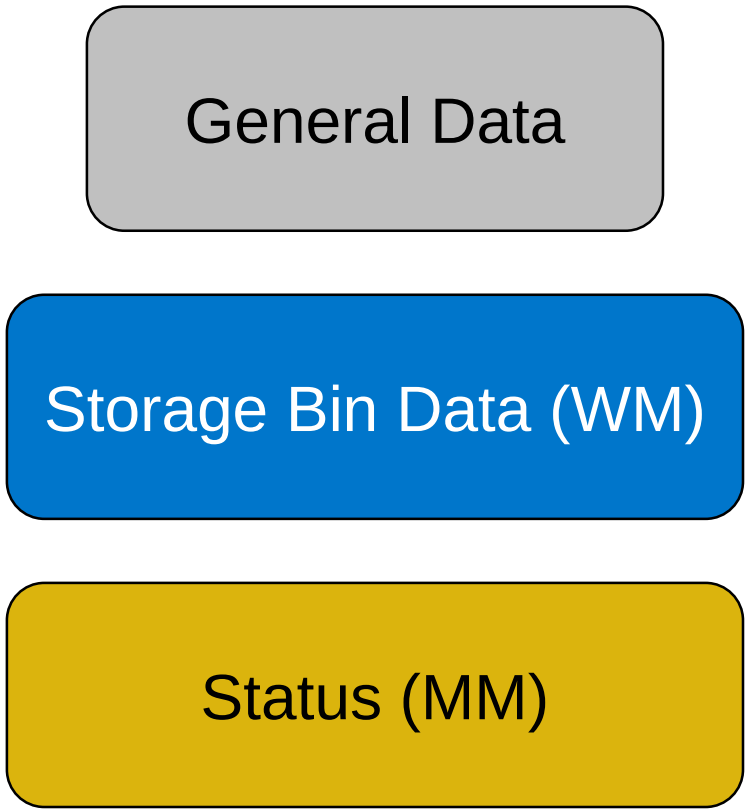
Storage Bin Master Data

- Storage Bin: Smallest addressable unit in a warehouse

  - General Data
    - Warehouse
    - Storage type
    - Storage bin number

  - Storage Bin Data
    - Storage section
    - Picking area
    - Fire-containment section
    - Bin type
    - Max. weight
    - Total capacity

  - Status
    - Blocked for putaway
    - Stock removal block
    - Blocking reason

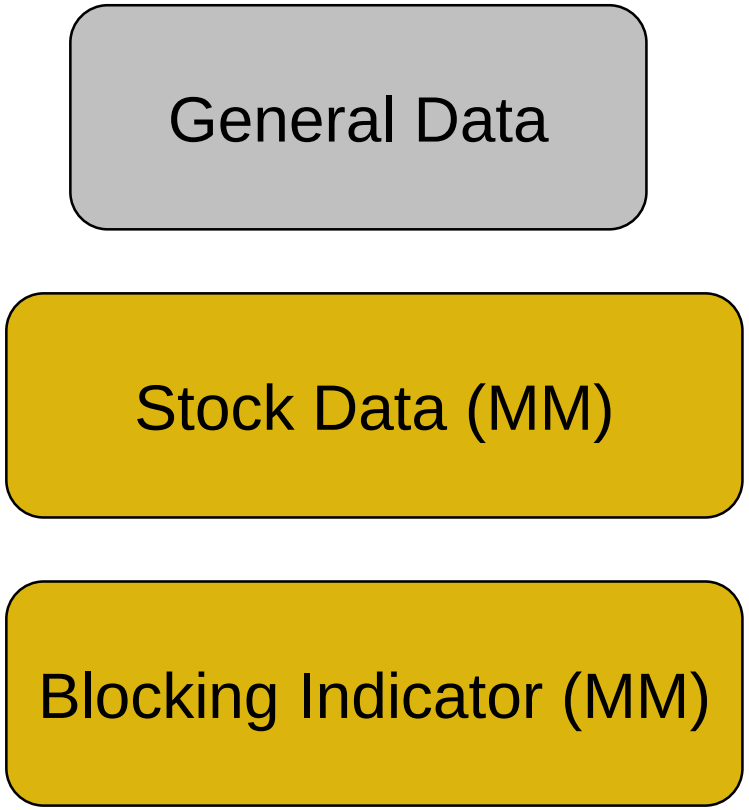


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Storage Bin Master Data

- Quant: Smallest addressable unit of measure for a material in SAP S/4HANA

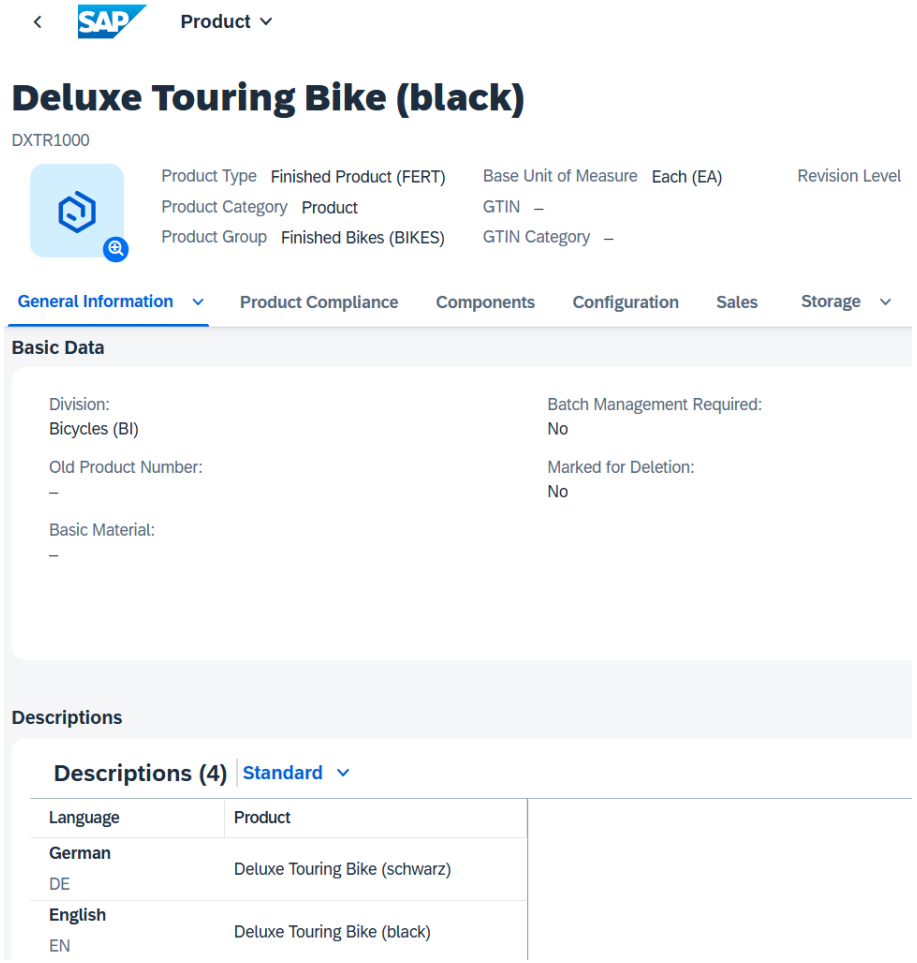
  - General Data
    - Material
    - Plant / Storage location
    - Batch
    - Warehouse
    - Storage type
    - Storage bin
  - Stock Data
    - Total stock
    - Available stock
  - Block Indicator
    - Block
    - Putaway/removal status



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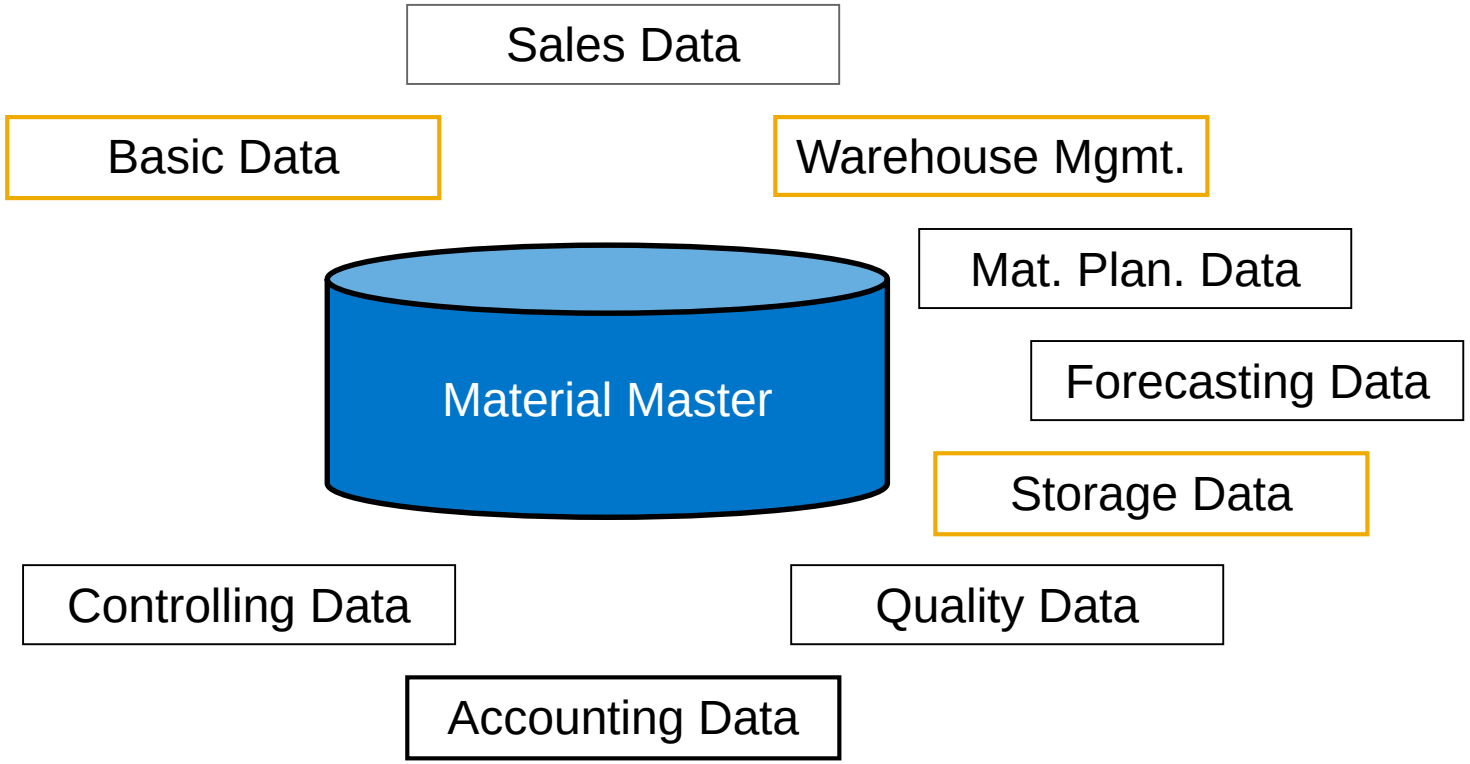
Material Master Data

- Material Master
  - Contains all the information a company needs to manage a material
  - It is used by most components within the SAP system
    - Sales and Distribution
    - Materials Management
    - Production
    - Plant Maintenance
    - Accounting/Controlling
    - Quality Management
  - Material master data is stored in functional segments called Views



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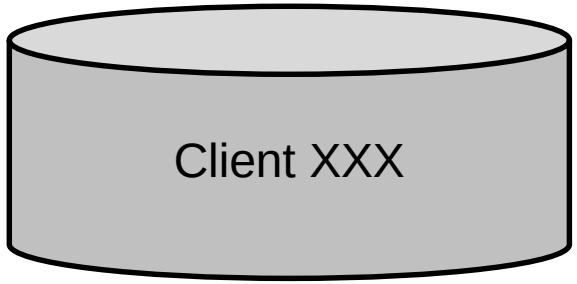
Material Master Views



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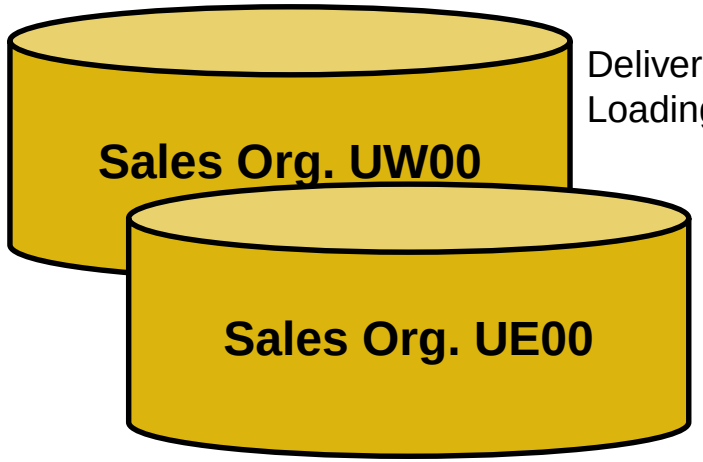
Material Master

General Information relevant for the entire organization:



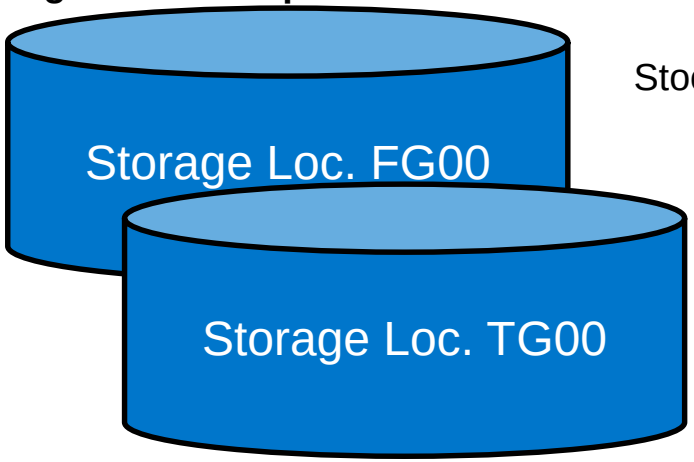
Name
Weight
Unit of Measure

Sales specific information:



Delivering Plant
Loading Grp

Storage Location specific information:

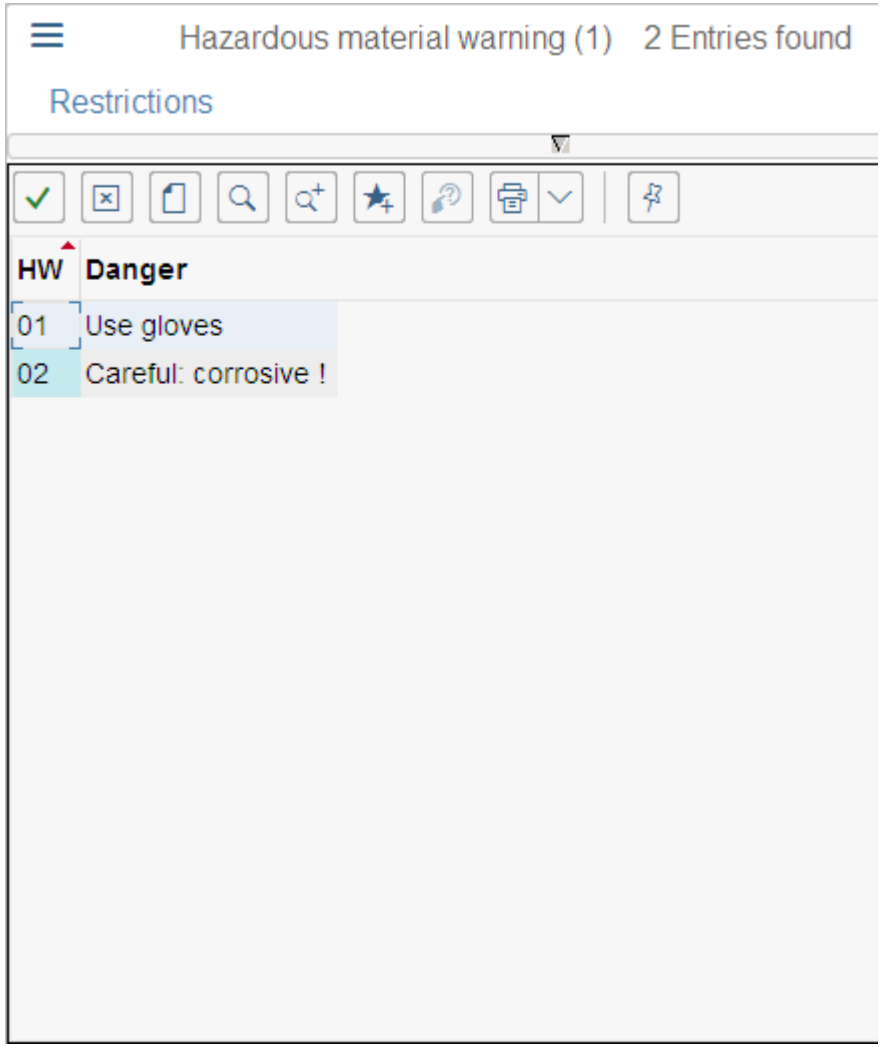


Stock Qty

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Hazard Master Data

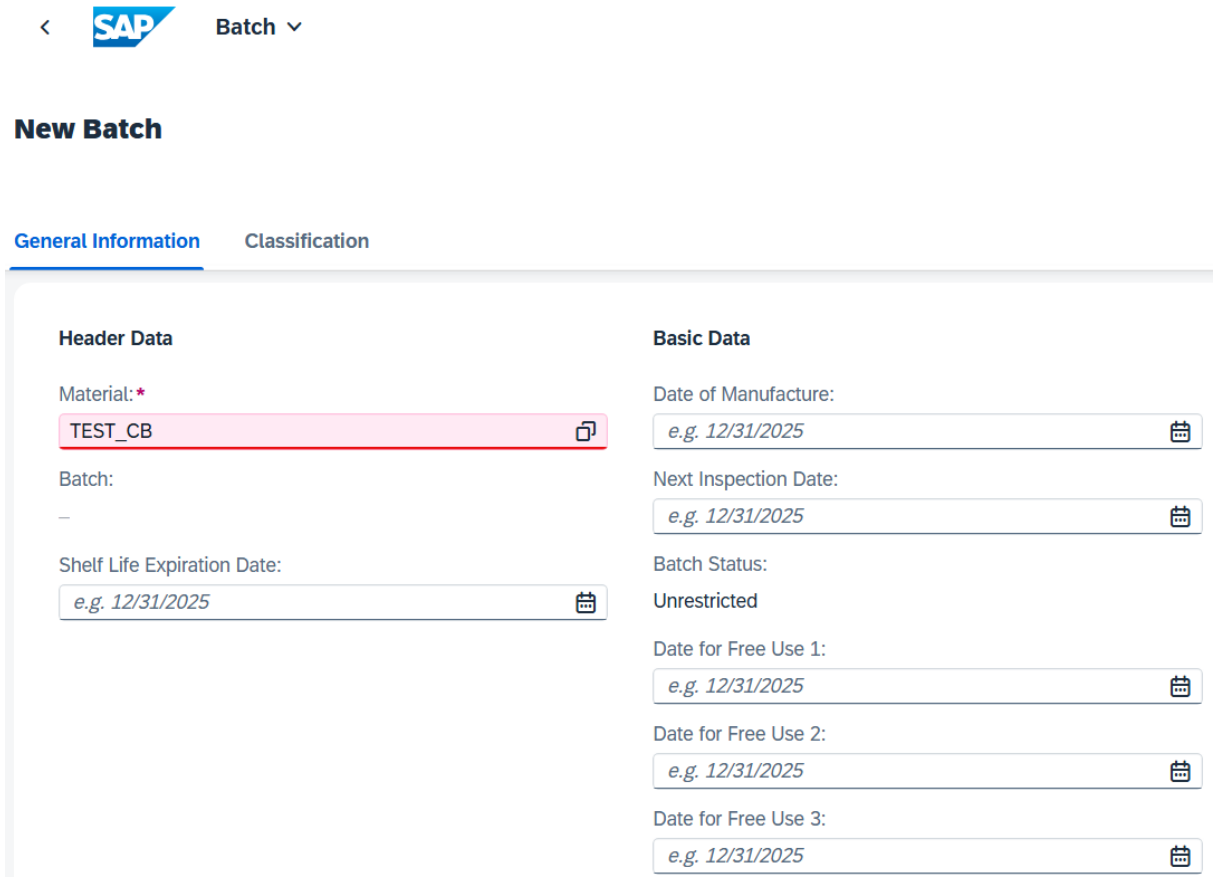
- Storage area for hazardous materials
  - Flammable liquids
  - Toxic materials
  - Radioactive materials
- General Data
  - Storage class
  - Water pollution class
  - Flash point
  - Aggregate state
  - Hazardous material warning
  - Hazardous substance number



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Batch Master Data

- Subset of Material
  - homogeneous series of unreproducible units with a unique specification
- Basic Data 1 & 2
  - Expiration date
  - Inspection date
  - Trading data
  - Administration data
  - Text data
- Classification of Batch
- Material data



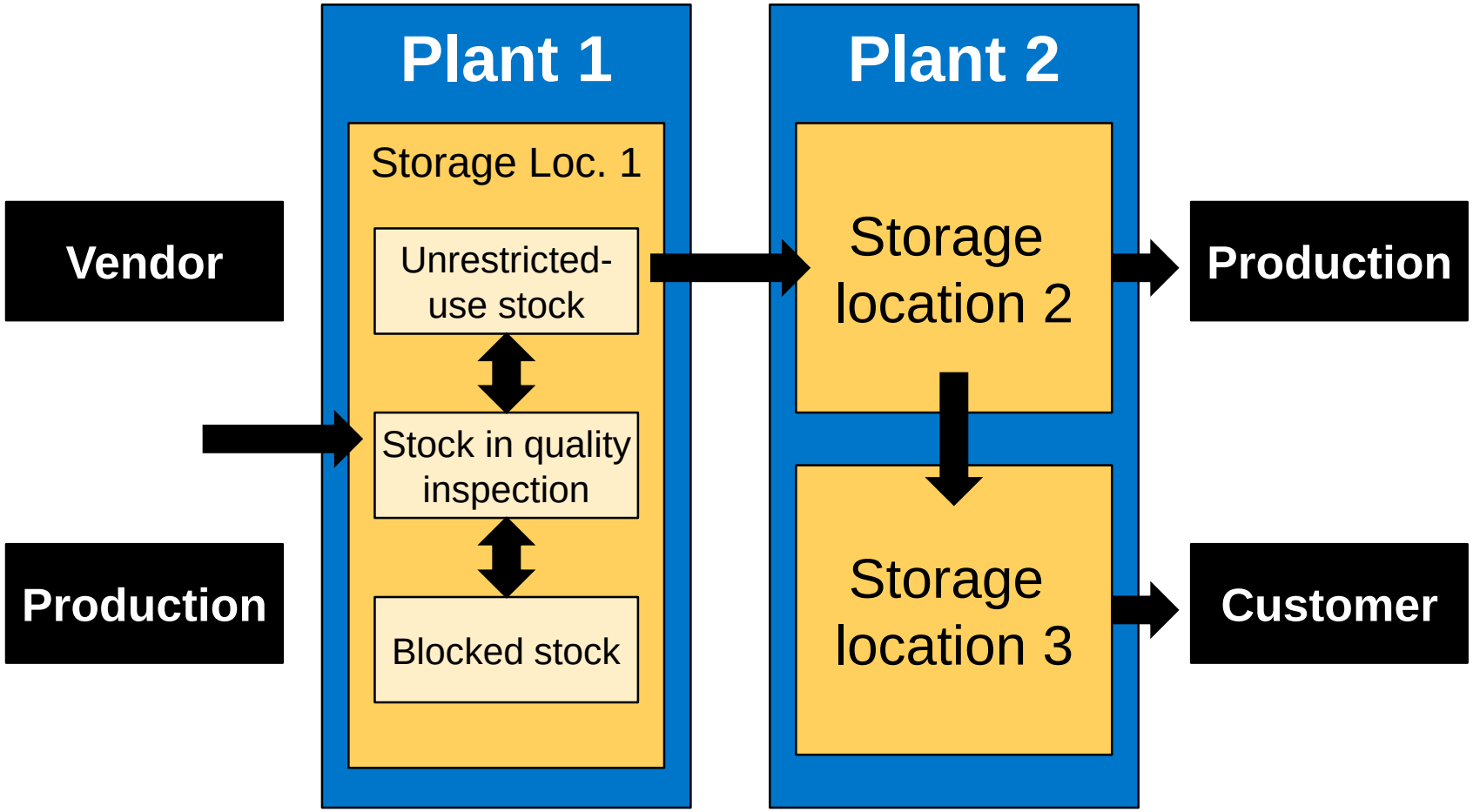
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Agenda

- WM Organizational Structure
- WM Master Data
- WM Process Management and Control

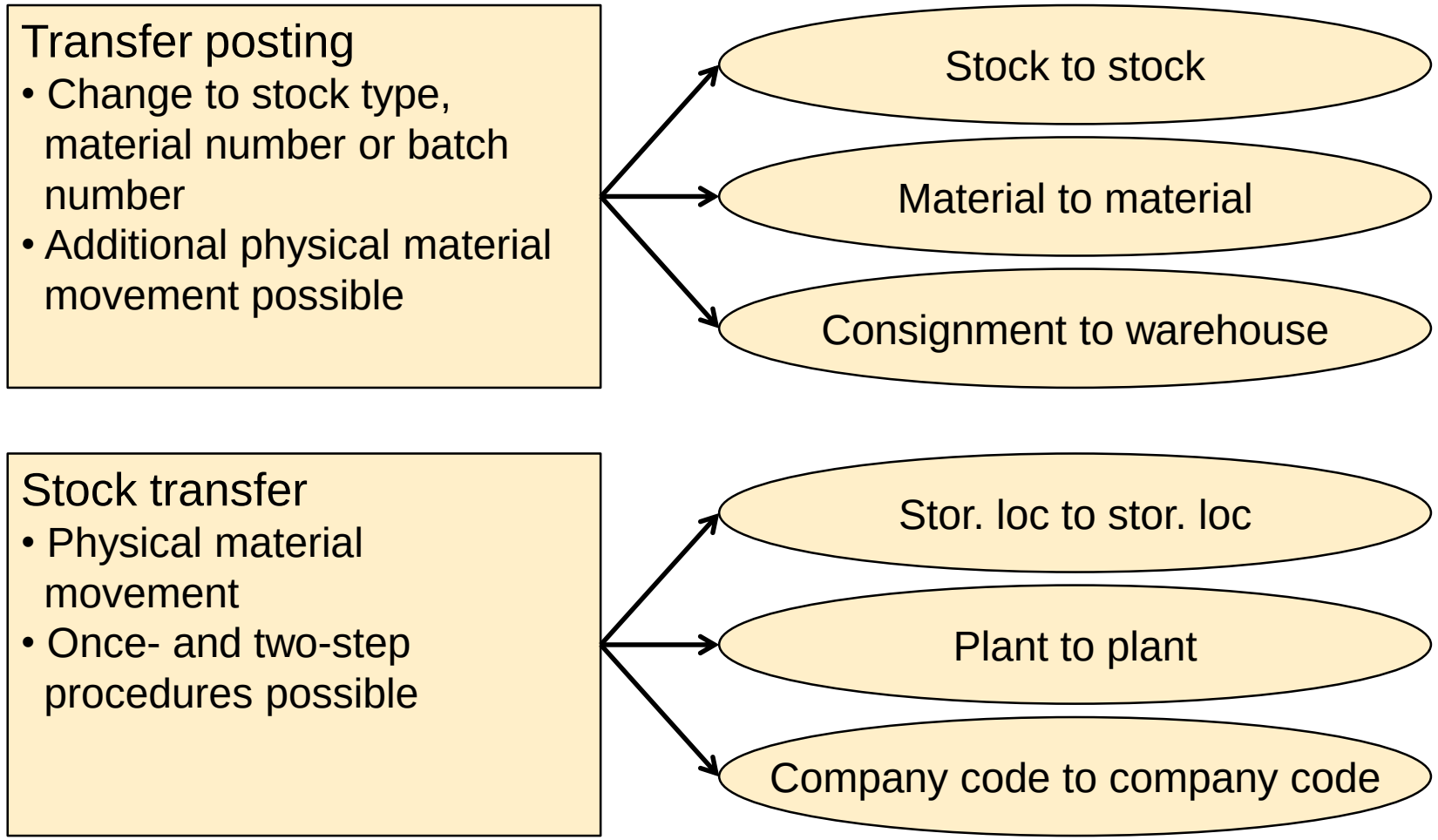
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WM Types of goods movements



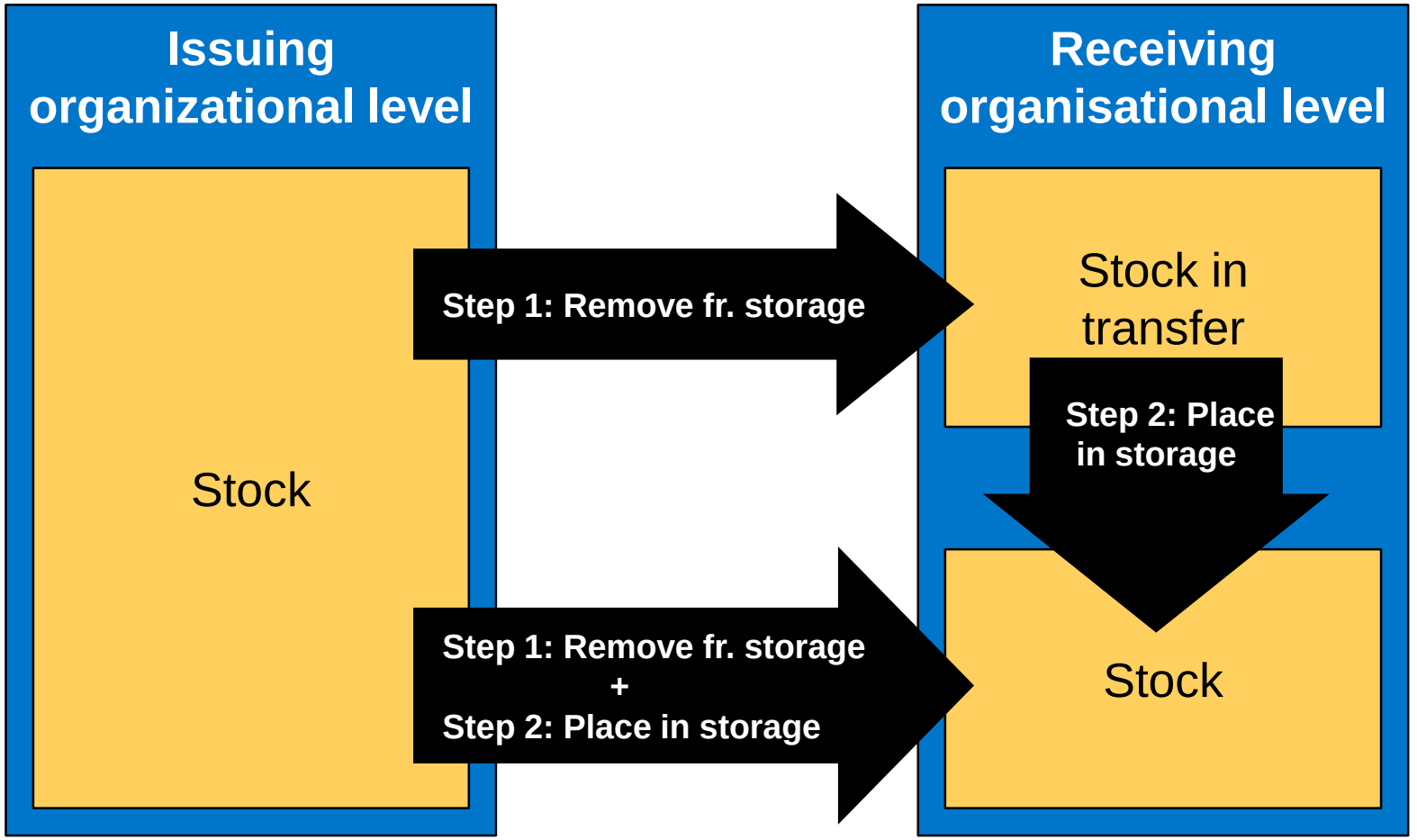
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Transfer posting and stock transfer



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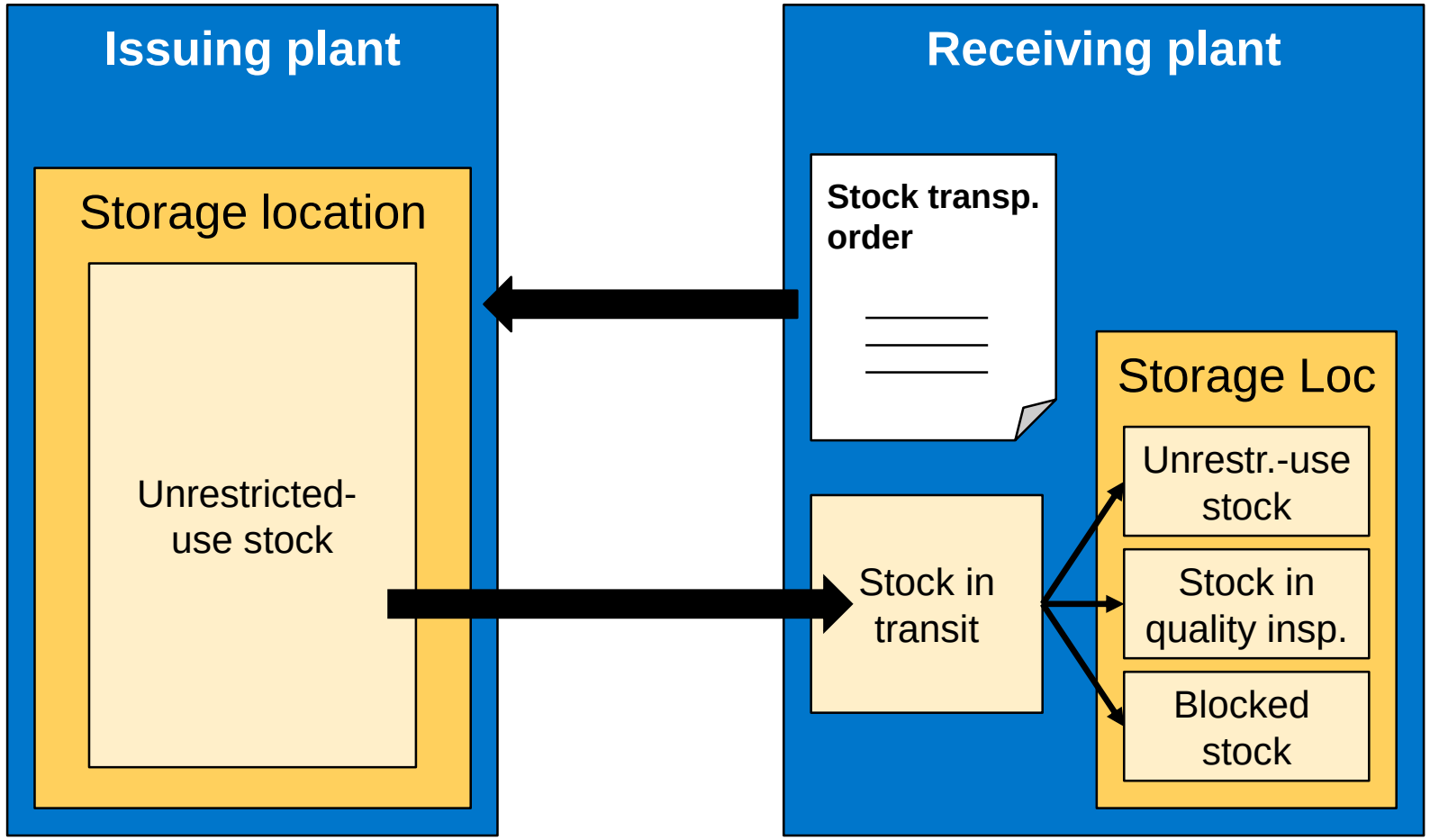
One-Step/Two-Step Procedures



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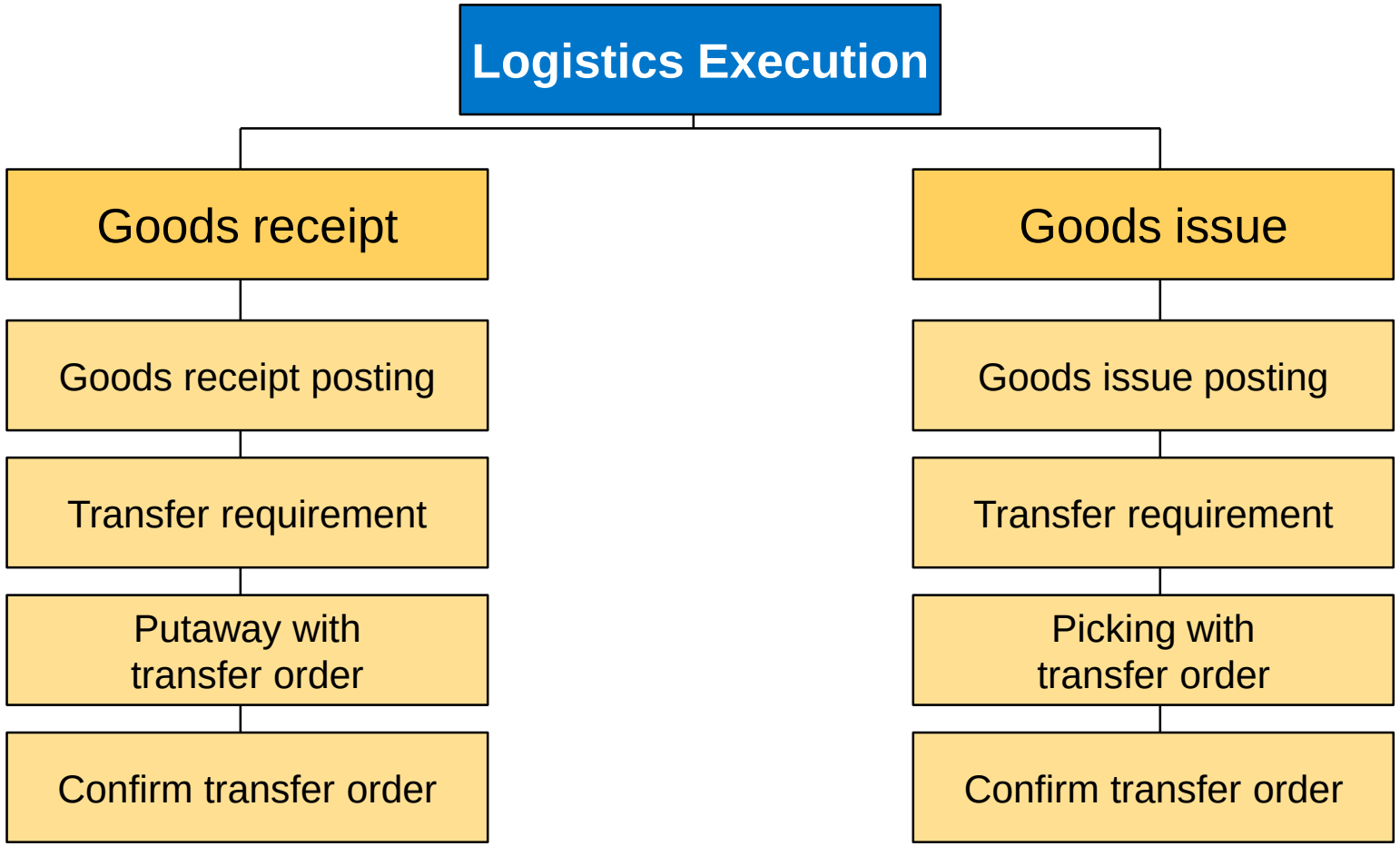
Stock transport order

- A Stock Transport Order (STO) is a purchase order used to request or instruct a plant to transport material from one plant to another (that is, to effect a long distance physical stock transfer) within the same corporate enterprise.
- The stock transport order allows delivery costs incurred as a result of the stock transfer to be charged to the material transported.



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WM Processes



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WM Process Management and Control

- Posting change notice (PCN)
  - Generated by posting change processes in WM-administrated stocks
  - Functionality similar to transfer requests
  - Not necessarily combined with a physical goods movement

- Transfer requirement (TR)
  - A request to transfer materials at a particular time from a source storage bin to a destination storage bin in a warehouse complex
  - Represents the expected and scheduled goods movements in WM
  - Normally generated by postings in the inventory management
  - Consists of transfer request header and line items

- Transfer order (TO)
  - Central documents for WM
  - Every material movement requires a transfer order
  - No difference between real and logical movement

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Transfer Order

- Creation of transfer order depends on preceding document
  - Transfer request
  - Posting change notice
  - Outbound delivery
  - Inbound delivery

- Types of Transfer Order Creation
  - Manual (TR, PCN, material document)
  - Direct TO-Creation
  - Automatic TO-Creation
  - Manual (Delivery Monitor)

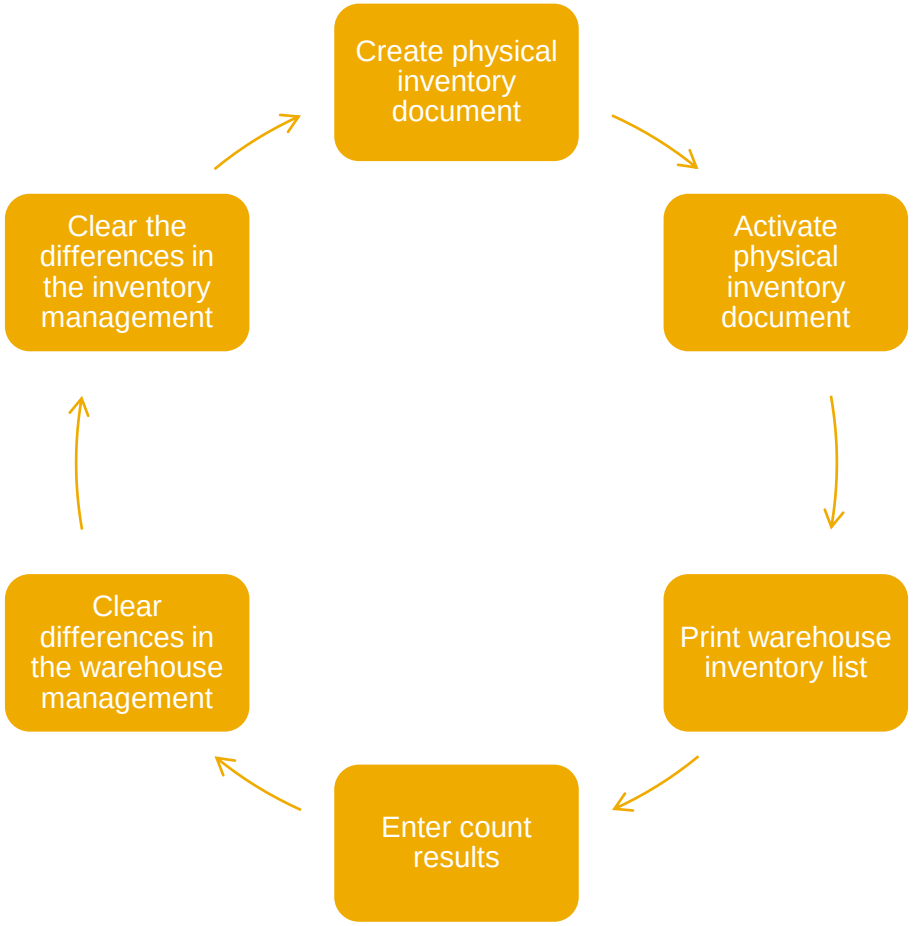
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Inventory

- Periodic inventory
- Continuous inventory
- Inventory on putaway
- Zero stock check
- Sample-based physical inventory
- Cycle-Counting

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Physical inventory process



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